

Use the analysis to plan a measured pilot

Understand customer groups. Improve sales planning. Keep decisions under review.

CUSTOMER INSIGHT

2 groups

Broad patterns to investigate

FORECAST READINESS

Limited

Simple rules still perform better

FINANCIAL IMPACT

Unproven

ROI requires a controlled pilot

Historical Olist study: findings support investigation, not a promise of live commercial results.

Two business decisions guide the project

CUSTOMER PLANNING

Who needs attention?

Use purchase patterns to design and test relevant outreach.

OPERATIONS PLANNING

What may sell next?

Compare forecasts with simple rules before using them to order stock.

Scope: a 6,000-customer sample and two training-selected five-product sets. Recorded sales are not total demand.

The large single-order group still matters

SINGLE-ORDER GROUP

5,800 customers

Median recorded spend: BRL 84.99
Share of sample spending: 93.8%

REPEAT / BROADER BASKET

200 customers

Median recorded spend: BRL 176.95
Share of sample spending: 6.2%

A smaller individual basket does not make the largest customer group unimportant.

These are sample profiles, not validated lifetime-value or loyalty categories.

Investigate the first-purchase experience

INVESTIGATE

Experience

Review delivery delays, ratings and product issues.

DESIGN

Relevant outreach

Include single-order customers; avoid segment-only exclusion.

EVALUATE

Incremental benefit

Compare with a randomized control group over the same period.

Repeat-purchase classification was retired. One observed order does not mean permanent churn.

Keep simple forecasting rules in the decision

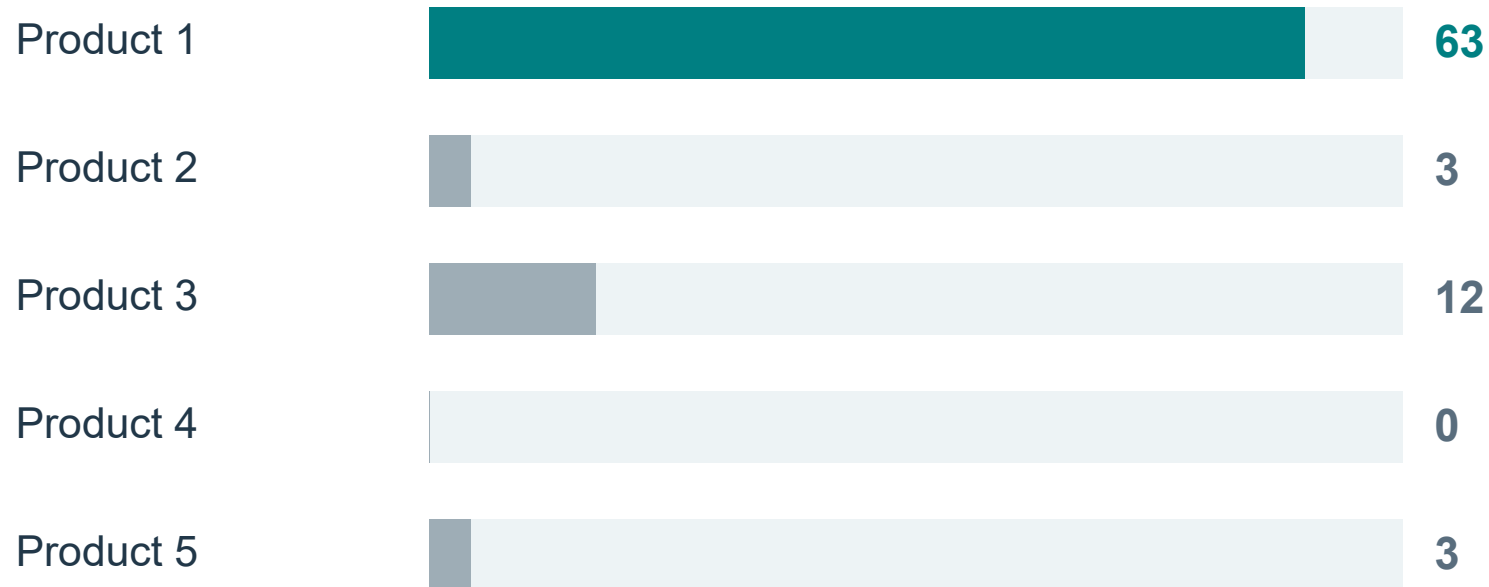


Weekly RF error is 14.2% higher. Daily models matched 0/5 observed trend directions.

Error is average items per product-week, across four horizons. Human review remains necessary.

Concentrate review where recorded activity is

Recorded items across 16 evaluation weeks



78%

of this small product set's recorded sales came from Product 1. Review its forecasts first.

For quiet products, check availability before assuming declining demand or changing stock orders.

What would a customer pilot need to earn?

ASSUMPTIONS: 1,000 customers; BRL 40 contribution per extra order; BRL 1 contact cost + BRL 500 setup.

Assumed uplift	Extra orders	Net value (BRL)	ROI
1 percentage point	10	-1,100	-73.3%
3 percentage points	30	-300	-20.0%
5 percentage points	50	+500	+33.3%

Break-even: +3.75 percentage points, or at least 38 extra orders. These are scenarios, not predictions.

Keep customer access and uncertainty visible

UNEQUAL SAMPLE COVERAGE

3.95 pp

State selection-rate gap for a hypothetical offer

IMPORTANT MISSING DATA

4 attributes

No gender, race, age or income data

PRACTICAL SAFEGUARD

Human review

Avoid automatic exclusion or replenishment

The state gap warrants investigation; small groups make it uncertain.

Proposed mitigations need testing. Geography is not a substitute for protected attributes.

Run a small pilot with clear success measures

Stage	Action	Evidence needed
Prepare	Confirm costs, consent and availability	Usable data; named owners
Test customers	Randomize outreach and control	Incremental contribution; reach
Test forecasts	Compare on new dates in shadow mode	Error vs simple benchmark
Decide	Review value, uncertainty and coverage	Documented go / revise / stop

Use newly collected, relevant data. Historical 2017-2018 results do not establish current performance.

Recommended next step: a measured pilot

- 1 Keep the large single-order group in scope.**
- 2 Retain simple sales benchmarks and human review.**
- 3 Invest in better evidence before automation.**

Success means positive incremental contribution, better forecast error on new dates, and acceptable customer coverage with uncertainty reported.

No proven ROI yet. Report complete; fresh Windows analysis run verified. Fairness gaps remain.